

Student Name:	Start Date:	End Date:
Course : Flutter App Development		
Trainer Name:		

Lecture Session Details (Basic)	Trainer Sign	Student Sign	Management Sign
Session 1: Introduction to Programming Fundamentals • What is programming? • Importance in modern technology • Overview of languages & logic building			
Session 2: Introduction to Dart Programming • What is Dart? • Features of Dart • Installing Dart & setup			
Session 3: Variables & Data Types • Variables concept • var, dynamic, final, const • Naming rules			
Session 4: Data Types in Dart • int, double • String, bool • Type conversion			
Session 5: Operators in Dart • Arithmetic operators • Relational & logical operators • Assignment operators			
Session 6: Collections – List • List creation • Access & modify elements • Common list methods			
Session 7: Collections – Set & Map • Set (unique values) • Map (key-value pairs) • Practical examples			

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Session 8: Conditional Statements • if, else if, else • Nested conditions • Real-life examples			
Session 9: Loops in Dart • for loop • while loop • break & continue			
Session 10: Functions Basics • Function definition • Parameters & return values • Code reuse concept			
Session 11: Arrow Functions & Optional Parameters • Arrow syntax • Named & optional parameters • Default values			
Session 12: OOP – Classes & Objects • Class structure • Creating objects • Real-world examples			
Session 13: Constructors & Inheritance • Default & parameterized constructors • Inheritance concept • Code reuse			
Session 14: Encapsulation & Exception Handling • Data hiding • try-catch • Handling runtime errors			
Session 15: Asynchronous Programming • Future concept • async & await • Real-world use (API calls)			
Session 16: Introduction to Flutter • What is Flutter? • Advantages • Cross-platform concept			

Lecture Session Details (Basic)	Trainer Sign	Student Sign	Management Sign
Session 17: Flutter SDK & Setup • Installing Flutter • IDE setup • Running first app			
Session 18: Flutter Project Structure • main.dart file • folders overview • pubspec.yaml			
Session 19: What are Widgets? • Everything is widget concept • Widget tree • UI building basics			
Session 20: Stateless Widgets • Definition • Use cases • Example UI			
Session 21: Stateful Widgets • State concept • Lifecycle basics • Updating UI dynamically			
Session 22: Basic Widgets – Text, Image, Icon • Display text • Adding images • Using icons			
Session 23: Layout Widgets – Container • Styling UI • Padding, margin • Decoration			
Session 24: Row & Column • Horizontal & vertical layout • Alignment properties • Nested layouts			
Session 25: ListView • Scrollable lists • Builder method • Dynamic data display			

Lecture Session Details (Basic)	Trainer Sign	Student Sign	Management Sign
Session 26: GridView • Grid layout • GridView.count & builder • UI design examples			
Session 27: Scaffold & AppBar • Basic app structure • AppBar design • Floating button			
Session 28: Navigation Basics • Moving between screens • Navigator.push & pop • Passing data			
Session 29: Material Design Principles • UI guidelines • Colors & typography • User experience basics			

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Session 30: Flexible & Expanded Widgets • Space management • Flex behaviour • Responsive layouts			
Session 31: Themes & Styling • ThemeData • Colors & fonts • App consistency			
Session 32: Creating Custom Widgets • Reusable UI components • Code optimization • Best practices			
Session 33: State Management – setState • State basics • Updating UI • Simple examples			

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Session 34: Provider (State Management) • Introduction to Provider • Managing global state • Practical use case			
Session 35: Introduction to API • What is API? • Real-world examples • Client-server model			
Session 36: REST API Basics • GET, POST, PUT, DELETE • API structure • Endpoints			
Session 37: Data Storage Concepts • Local vs cloud storage • Data persistence basics			
Session 38: SharedPreferences • Storing simple data • Key-value storage • Login example			
Lecture Session Details	Trainer Sign	Student Sign	Center Head Sign
Session 39: Introduction to Database & SQLite • What is Database? • Types of Databases • What is SQLite? • Features and Advantages of SQLite			
Session 40: SQLite Installation & Database Basics • SQLite Environment Setup • Creating Database • SQLite Commands • Database Files			
Session 41: SQLite Data Types • INTEGER • REAL • TEXT • BLOB • NULL			

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Session 42: Creating Tables • CREATE TABLE Statement • Column Definitions • Table Structure Design			
Session 43: Constraints in SQLite • PRIMARY KEY • FOREIGN KEY • UNIQUE • NOT NULL • DEFAULT			
Session 44: CRUD Operations – INSERT & SELECT • INSERT Statement • SELECT Statement • Retrieving Records • Column Aliases			
Session 45: CRUD Operations – UPDATE & DELETE • UPDATE Statement • DELETE Statement • WHERE Clause • Practical Examples			
Session 46: Operators in SQLite • Arithmetic Operators • Comparison Operators • Logical Operators • Special Operators			
Session 47: Clauses in SQLite • WHERE Clause • ORDER BY Clause • GROUP BY Clause • HAVING Clause • LIMIT Clause			
Session 48: Functions in SQLite • Aggregate Functions • COUNT(), SUM(), AVG() • MIN(), MAX() • String Functions			

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Session 49: Joins in SQLite • INNER JOIN • LEFT JOIN • CROSS JOIN • Practical Examples			
Session 50: Subqueries & Views • Nested Queries • Scalar Subqueries • Creating Views • Managing Views			
Session 51: Indexes & Performance • What is Index? • Creating Indexes • Advantages of Indexes • Query Optimization Basics			
Session 52: Transactions in SQLite • BEGIN Transaction • COMMIT • ROLLBACK • ACID Properties			
Session 53: Mini Database Project • Student Management Database • Table Creation • CRUD Operations • Joins & Reports			